UNIT-2

A] ARITHMETIC MEAN

* For ungrouped data:-

- Population $\Rightarrow \mu = \frac{\Sigma x}{N}$

- Sample $\Rightarrow \overline{X} = \frac{\Sigma x}{n}$

* For grouped data:-

Given: Class Interval	Given: freq (f)	Mid Pt (x)	(f×x)
0 - 49.99	78	25	1950
50 - 99.99	123	75	9225
100 - 149.99	187	125	23375
150 - 199.99	82	175	14350
200 - 249.99	51	225	11475
250 - 299.99	47	275	12925
300 - 349.99	13	325	4225
350 - 399.99	9	375	3375
400 - 449.99	6	425	2550
450 - 499.99	4	475	1900
	$\Sigma f = 600$		$\Sigma = 85350$

$$\text{Mean} = \frac{\Sigma(f \times x)}{\Sigma f} = \frac{85350}{600} = 142.25$$

* Coding Method :- ☆ h(Class Interval) = 8

☆ A(Assumed Mean) = 28 (Mid Value is assumed depending on your convenience)

Given Class Interval	Given Frequency (f)	Mid Value (x)	☆ $d = \frac{x-A}{h}$
0 - 8	8	4	-3
8 - 16	7	12	-2
16 - 24	16	20	-1
24 - 32	24	28	0
32 - 40	15	36	1
40 - 48	7	44	2

$N = \Sigma f = 77$

fd
-24
-14
-16
0
15
14

$\Sigma fd = -25$

$$\bar{x} = A + \frac{h\Sigma(fd)}{N}$$

$$= 28 + \frac{8(-25)}{77}$$

$$= 25.404$$

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B] WEIGHTED MEAN

Eg] A store advertises (If our average prices are not equal or lower than everyone else, you get it for free).
One of the customer came into the store and discussed about the bills for six items, she bought from the competitors for an average price than this store.

Customer item cost

\$ 1.29	2.97	3.49	5	7	10.95

Store price

\$ 1.35	2.89	3.19	4.98	7.59	11.50

The store told the customer, my ad refers to a WEIGHTED AVERAGE of these items. Our average is lower because our sales of these items have been.

Sales

7	9	12	8	6	3	⇒ 45

Will the store get in trouble?

Soln] Unweighted Average

$\bar{x}_c = \$5.11$
$\bar{x}_\$ = \$5.25$

Weighted Average

$$\bar{x}_c = \frac{\Sigma(W \times x)}{\Sigma W}$$

$$\bar{x}_c = \frac{7 \times 1.29 + 9 \times 2.97 + 3.49 \times 12 + 5 \times 8 + 7 \times 6 + 10.95 \times 3}{45}$$

$\bar{x}_c = 4.27$

$$\bar{x}_s = \frac{1.35\times7 + 2.89\times9 + 3.19\times12 + 4.98\times8 + 7.59\times6 + 11.5\times3}{45}$$

$= 4.3$

$\because \bar{x}_c < \bar{x}_s$

$\therefore$ The store will get in trouble.

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C] GEOMETRIC MEAN

Eg] 100 \$ are deposited in savings account for 5 years with below given interest rate. Compare AM & GM.

Year	Interest Rate (%)	Growth Factor	Savings at end of each year
1	7	1.07	\$ 107
2	8	1.081	\$ 115.56
3	10	1.10	\$ 127.12
4	12	1.12	\$ 142.37
5	18	1.18	\$ 167.99

Simple Arithmetic Mean (Interest) G.F = 1.11 {11% per year}

$100 × 1.11 × 1.11 × 1.11 × 1.11 × 1.11 = 168.505 \$

$GM = \sqrt[5]{1.07 \times 1.08 \times 1.10 \times 1.12 \times 1.18}$
$= 1.1093 \Rightarrow$ avg. G.F

10.93% per year

* MEDIAN

Eg]

Class ($)	Frequency	C.F
0-49.99	78	78
50-99.99	123	201
100-149.99	187	388 → Median Class
150-199.99	82	470
200-249.99	51	521
250-299.99	47	568
300-349.99	13	581
350-399.99	9	590
400-449.99	6	596
450-499.99	4	600

$n = \Sigma f = 600$

$\text{Median} = \left(\frac{n+1}{2}\right)^{th} = \frac{600+1}{2} = 300.5$

∴ 300th & 301st item ∴ Median Class.

300 − 201 = 99th position
301 − 201 = 100th position

$\text{Width of Step} = \frac{\overbrace{150-100}^{\text{Median class ka top \& bottom ka difference.}}}{\underbrace{187}_{\text{freq of median class}}} = 0.267\ \$$

For 300^{th} item $= 100 + 98 \times 0.267$
$= 126.166$ \$

For 301^{st} item $= 126.17 + 0.267 = 126.433$ \$

$$\text{Median} = \frac{126.17 + 126.43}{2} = \underline{\underline{126.3}} \ \$$$

(Median cuz we needed 300.5^{th} position ka thing)

Formula for Median $\Rightarrow \dfrac{\left[\left(\frac{n+1}{2}\right) - (F+1)\right] W}{f_m} + L_m$

here,
F = sum of all class frequency upto but not inc Median Class
f_m = freq. of median class
W = Width of class interval
L_m = Lower limit of median class.

$$\therefore \text{Median} \Rightarrow \frac{\left[\left(\frac{600+1}{2}\right) - (201+1)\right] 50}{187} + 100$$

$\Rightarrow \underline{\underline{126.336 \ \$}}$

* MODE

$$Mo = LMo + \left(\frac{d_1}{d_1+d_2}\right) W$$

here,

d_1 = freq of Modal Class − freq of the class above it.

d_2 = freq of Modal Class − freq of the class below it.

LMo = lower limit of modal class

W = Width of modal class

$$Mo = 100 + \left(\frac{64}{169}\right) \times 50$$

$$= 118.93\$$$

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* ~~DATA VISUALISATION~~

UNIT 3 – DESCRIPTIVE STATISTICS

* POPULATION VARIANCE

$$\sigma^2 = \frac{\Sigma(x-\mu)^2}{N} = \frac{\Sigma x^2}{N} - \mu^2$$

where,
σ^2 = population variance
x = item (or) observation
μ = population mean
N = Total no. of items in the population

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* POPULATION VARIANCE & STANDARD DEVIATION

☆ For ungrouped data.

$\sigma = \sqrt{(\sigma^2)}$ upar wala (σ^2) lai yaha cuz same-same

☆ For grouped data.

$$\sigma^2 = \frac{\Sigma f(x-\mu)^2}{N} = \frac{\Sigma fx^2}{N} - \mu^2$$

* SAMPLE VARIANCE & STANDARD DEVIATION

☆ For ungrouped data.

$$S^2 = \frac{\Sigma(x-\bar{x})^2}{n-1} = \frac{\Sigma x^2}{n-1} - \frac{n\bar{x}^2}{n-1}$$

$S = \sqrt{S^2}$

★ For grouped data

$$S^2 = \frac{\sum f(x_i-\bar{x})^2}{n-1} \qquad S=\sqrt{S^2}$$

Eg] If we have a population of 15 vials, of compound produced in 1 day and we test each vial to determine its purity. Our data will look as below. Calculate the variance & standard deviation.

$$\mu = \frac{\sum x}{N} = \frac{2.49}{15} = 0.16$$

x Obsv	Mean μ	$x-\mu$	$(x-\mu)^2$
0.04	0.166	−0.126	0.0158
0.06	0.166	−0.106	0.011236
0.12		−0.046	~~0.0~~ 2.116×10^{-3}
0.14		−0.026	6.76×10^{-4}
0.14		−0.026	6.76×10^{-4}
0.15		−0.016	2.56×10^{-4}
0.17		0.004	1.6×10^{-5}
0.17		0.004	1.6×10^{-5}
0.18		0.014	1.96×10^{-4}
0.19		0.024	5.76×10^{-4}
0.21		0.044	0.1936 1.93×10^{-3}
0.21		0.044	~~0.1936~~ 1.93×10^{-3}
0.22		0.054	~~0.2916~~ 2.91×10^{-3}
0.24		0.074	5.476×10^{-3}
0.25		0.084	7.056×10^{-3}

$\sum(x-\mu)^2 = 0.0508$

$$\sigma^2 = \frac{\sum(x-\mu)^2}{N} = \frac{0.0508}{15} = 0.0034 \qquad \sigma = \sqrt{\sigma^2} = 0.0583$$

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Eg] The Vice President of a marketing team of fast food chain is studying the sales performance of 100 stores in Eastern district & has compiled this frequency distribution of annual sales. Calculate the variance & standard deviation.

($) Sales (000s)	freq (f)	Midpt (x)	fx	$x-\mu$
700 - 799	4	750	3000	-500
800 - 899	7	850	5950	-400
900 - 999	8	950	7600	-300
1000 - 1099	10	1050	10500	-200
1100 - 1199	12	1150	13800	-100
1200 - 1299	17	1250	21250	0
1300 - 1399	13	1350	17550	100
1400 - 1499	10	1450	14500	200
1500 - 1599	9	1550	13950	300
1600 - 1699	7	1650	11550	400
1700 - 1799	2	1750	3500	500
1800 - 1899	1	1850	1850	600
	N=100		$\Sigma fx = 125000$	

$$\mu = \frac{\Sigma f(x)}{N} = \frac{125000}{100} = 1250\$$$

$(x-\mu)^2$	$f(x-\mu)^2$
250000	1000000
160000	1120000
90000	720000
40000	400000
10000	120000
0	0
10000	130000
40000	400000
90000	810000
160000	1120000
250000	500000
360000	360000
	$\Sigma f(x-\mu)^2 = 6680000$

$$\sigma^2 = \frac{\Sigma f(x-\mu)^2}{N} = \frac{6680000}{100} = 66800$$

$$\sigma = \sqrt{\sigma^2} = 258.456$$

Eg] These data are the sample of daily production of fibre glass boats from a Miami manufacturer.

$x =$ 17 21 18 27 17 21 20 22 18 23

The company production manager feels that a standard deviation of more than 3 boats a day indicates unacceptable production rate variation. Should the manager be concerned about plant production rate.

Soln] $\bar{x} = \frac{\sum x}{n} = \frac{204}{10} = 20.4$

Sample Variance (σ^2) $= \frac{\sum (x-\bar{x})^2}{n-1}$

$$= \frac{(17-20.4)^2 + (21-20.4)^2 + (18-20.4)^2 + (27-20.4)^2 + (17-20.4)^2 + (21-20.4)^2 + (20-20.4)^2 + (22-20.4)^2 + (18-20.4)^2 + (23-20.4)^2}{9}$$

$$= \frac{11.56 + 0.36 + 5.76 + 43.56 + 11.56 + 0.36 + 0.16 + 2.56 + 5.76 + 6.76}{9}$$

$$= \frac{88.4}{9} = 9.8222$$

$\therefore$ Standard Deviation $(\sigma) = \sqrt{\sigma^2} = \sqrt{9.822}$

$\sigma = 3.134$

Yes, the manager should be concerned about plant production because sample standard deviation is 3.13 is more than the acceptable value of 3.

Eg) In an attempt to estimate potential future demand, The National Motor Company did a study asking married couples how many cars the average energy-minded families should own in 2024. For each couple national averaged the husband's & wife's responses to get the overall couple response. The answers were tabulated. Calculate the variance & standard deviation.

$$S^2 = \frac{\sum f(x-\bar{x})^2}{n-1}$$

$$\bar{x} = \frac{\sum fx}{n}$$

No. of Cars (x)	freq (f)
0	2
0.5	14
1	23
1.5	7
2	4
2.5	2
	$n = 52$

fx	$x-\bar{x}$	$(x-\bar{x})^2$	~~$f(x-\bar{x})^2$~~
0	-1.029	1.058	~~2.116~~
7	-0.529	0.279	~~[illegible]~~
23	-0.029	8.41×10^{-4}	~~8.41×10^{-4}~~
10.5	0.471	0.221	~~0.3315~~
8	0.971	0.942	~~1.884~~
5	1.471	2.163	~~5.4075~~
$\sum fx = 53.5$			~~$\sum f(x-\bar{x})^2$~~

$$\therefore \bar{x} = \frac{53.5}{52}$$

$$= 1.0289$$

$f(x-\bar{x})^2$

2.1178
3.9186
0.0193
1.5526
3.7712
4.3276

$\Sigma = 15.7071$

$$\sigma^2 = \frac{\sum f(x-\bar{x})^2}{n-1} = \frac{15.7071}{9} = 0.308$$

$$\therefore \sigma = \sqrt{\sigma^2} = 0.555$$

☆ Karl Pearson's Coefficient of Skewness:-

↳ Using Mode ⇒ $\frac{\bar{x} - \text{Mode}}{s}$

↳ Using Median ⇒ $\frac{3(\bar{x} - \text{Median})}{s}$

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☆ Sample Mean ⇒ $\bar{x} = \frac{\sum fx}{n}$

☆ Sample Median ⇒ $\frac{l + \left(\frac{n}{2} - Fc\right) \times w}{f}$

or

$$l + \frac{h}{f}\left(\frac{N}{2} - c\right)$$ ☆

c = Cumulative frequency.

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☆ Sample Mode $\Rightarrow L + \left(\frac{f_m - f_1}{2f_m - f_1 - f_2}\right) \times w$

☆ Sample Variance $\Rightarrow S^2 = \frac{1}{n-1}\left(\Sigma f x^2 - \frac{(\Sigma f x)^2}{n}\right)$

Eg] Calculate Karl Pearson's coefficient of skewness for the following data:-

Size x	freq f	fx	fx^2
1	10	10	10
2	18	36	72
3	Mode 30	90	270
4	25	100	400
5	12	60	300
6	3	18	108
7	2	14	98
	$\Sigma f = 100$	$\Sigma fx = 328$	$\Sigma fx^2 = 1258$

$SK = \frac{M - M_0}{SD}$

$M = \frac{\Sigma f x}{N} = \frac{328}{100} = 3.28$

$SD = \sqrt{\frac{\Sigma f x^2}{n} - \left(\frac{\Sigma f x}{n}\right)^2} = \sqrt{\frac{1258}{100} - (3.28)^2} = 1.3496$

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$\therefore SK = \frac{3.28-3}{1.3496} = 0.2074$

Hence, the distribution is slightly positively skewed.

Eg] Calculate Karl Pearson's coefficient of skewness of the following data:-

Hourly Wages ($)	No. of Workers
40-50	5
50-60	6
60-70	8
70-80	10
80-90	25
90-100	30
100-110	36
110-120	50
120-130	60
130-140	70
	N=300

Since the max frequency that is 70 is at the end of frequency distribution, mode is ill-defined. Hence, skewness can be found with median.

Jab bhi class interval totally continuous hota hai toh $\frac{N}{2}$.

$\frac{N}{2} = \frac{300}{2} = 150$

$$Me = l + \frac{h}{f}\left(\frac{N}{2} - C\right)$$

Eg] For a distribution Karl Pearson coefficient of skewness is 0.64. Standard deviation is 13 and Mean is 59.2. Find Mode & Median.

~~soln] $\bar{x} = 59.2$~~

~~SD = 13~~

~~Sk = 0.64~~

~~$0.64 = \frac{(\text{Mode} - 59.2)}{13}$~~

~~Mode = 50.88~~

3(59.2 −

Eg] The no. of students absent in a class was recorded everyday for 60 days and the information is given in the following frequency distribution table. Find skewness.

No. of students absent (x)	No. of days (f)	fx	fx^2
0	3	0	0
1	6	6	6
2	18	36	72
3	18	54	162
4	8	32	128
5	5	25	125
6	2	12	72
	60	$\Sigma fx = 165$	$\Sigma fx^2 = 565$

cf
3
9
27
45
53
58
60

$\bar{x} = \frac{165}{60} = 2.75$

$\frac{N}{2} = \frac{60}{2} = 30$ ∴ 45

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Eg] The following table gives the distribution of weights of hundred babies at a certain hospital. Calculate Karl Pearson's Coefficient of skewness. (using mode here cuz highest value dekh raha & no repetition)

Weights (pounds)	No. of babies (f)	x	fx	fx^2
3-5	10	4	40	160
5-7	30	6	180	1080
7-9	28	8	224	1792
9-11	18	10	180	1800
11-13	14	12	168	2016
	$\Sigma f = 100$		$\Sigma fx = 792$	$\Sigma fx^2 = 6848$

UNIT-4 PROBABILITY

1) ~~Trial~~

* Axiomatic definition of Probability

Let E be a random experiment &
S be the sample space.
we, define the probability $P(A)$ for every element $A(S)$ satisfying the following axioms:-

1) $P(A) \geq 0$ (Axiom 1)
2) $P(S) = 1$ (Axiom 2)
3) $P(A \cup B) = P(A) + P(B)$ (Axiom 3)

If A & B are two exclusive events i.e. they are disjoint sets.

* Laws of Probability

1) $P(\phi) = 0$

2) $P(\bar{A}) = 1 - P(A)$

3) $P(B \cap \bar{A}) = P(B) - P(A \cap B)$

4) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

5) $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$

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Eg] What is the chance that a leap year selected at random will contain 53 sundays?

Soln] $P(A) = \frac{m}{n}$

52 week 2 days

S = {(M, Tu), (Tu, W), (W, Th), (Th, Fri), (Fri, Sat), (Sat, Sun), (Sun, M)}.

$\therefore \; 8 = \frac{2}{7}$

Eg] In a play of 2 dies a person loses is the sum obtained is 2 or 4 or 12. He wins if the sum is 5 or 11. Find the ratio of his probability of losing to the probability of winning in the first throw.

Soln] To Find $\Rightarrow \frac{P(\text{losing})}{P(\text{winning})}$

Eg] A bag contains 50 tickets numbered 1, 2, 3, ..., 50. Now 5 tickets are drawn at random and arranged in ascending order. What is the probability that $x_3 = 30$.

Soln] To Find: $P(A) = \frac{m}{n}$. $x_1 < x_2 < x_3 < x_4 < x_5$

x_1, x_2 ke possibilities will be 1–29 tak

x_4, x_5 ke possibilities will be 31–50 tak

$$= \frac{{}^{29}C_2 \cdot {}^{20}C_2}{{}^{50}C_5}$$

$$= 0.0364 = \frac{551}{15134}$$

Eg] There are 11 tickets in a box having numbers 1 to 11. 3 tickets are drawn one after the other, without replacement. Find the probability that they are drawn in the order of
i) Even Odd Even.
ii) Odd Odd Even.

6 odd, 5 even

Soln] $P(\text{even, odd, even}) = \frac{5}{11} \times \frac{6}{10} \times \frac{4}{9} = \frac{4}{33}$

$P(\text{odd, odd, even}) = \frac{6}{11} \times \frac{5}{10} \times \frac{5}{9} = \frac{5}{33}$

* Conditional Probability

- $P(A|B) = \frac{P(A \cap B)}{P(B)}$ • $P(B|A) = \frac{P(B \cap A)}{P(A)}$

* Multiplication Rule of Probability

1) Simultaneous happening of two events :- • $P(A|B)$
   • $P(B|A)$

2) More than 2 events (3), A_1, A_2, A_3 :-

$$P(A_1 \cap A_2 \cap A_3) = P(A_1) \cdot P(A_2|A_1) \cdot P(A_3|A_1 \cap A_2)$$

Eg] Suppose that in a general population, there are 51% men & 49% women and that the proportions of colour-blind men and women are shown in the table.

	Men	Women
Colorblind	0.04	0.002
Not-Colorblind	0.47	0.488
Total	0.51	0.49

i) If a person is drawn at random from this population & is found to be a man, what is the probability that the man is colorblind.

ii) What is the probability of being colorblind given that the person is a female.

i) $P(A|B) = \frac{0.04}{0.51} = 0.078$

ii) $P(A|B) = \frac{0.002}{0.49} = 0.00409$

Eg] Consider an experiment of tossing two fair dies. Consider two events A & B defined by A = sum of numbers on the dies is 6 or 8. B = atleast one dice shows four.

Soln] Sample space

$A = \{(1,5), (5,1), (2,4), (4,2), (3,3), (2,6), (6,2), (3,5), (5,3), (4,4)\}$

$B = \{(1,4), (2,4), (3,4), (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (5,4), (6,4)\}$.

$P(A) = \frac{10}{36}$, $P(B) = \frac{11}{36}$, $P(A \cap B) = \frac{3}{36}$

$\therefore P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{3}{11}$

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Eg] A box contains 4 bad & 6 good tubes. 2 are drawn out of the box at a time. One of them is tested & found to be good. What is the probability that the other one is also good?

Soln] A → first tube is good
B → second tube is good.

To Find: $P(B|A) = \dfrac{P(B \cap A)}{P(A)} = \dfrac{{}^6C_2 / {}^{10}C_2}{{}^6C_1 / {}^{10}C_1} = \dfrac{5}{9}$

Eg] A box of light bulbs contains 30 bulbs of which 5 bulbs are defective. If 3 of the bulbs are selected at random & taken out from the box in succession w/o replacement. Find the probability that all 3 bulbs are defective.

Soln] A → 1^{st} bulb is defective
B → 2^{nd} "
C → 3^{rd} "

$P(A \cap B \cap C) = P(A)\, P(B|A)\, P(C|B \cap A)$

$= \dfrac{5}{30} \times \dfrac{4}{29} \times \dfrac{3}{28}$

$= \dfrac{1}{406} = 0.00246.$

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i) $P(A|B) = \frac{0.04}{0.51} = 0.078$

ii) $P(A|B) = \frac{0.002}{0.49} = 0.00409$

Eg] Consider an experiment of tossing two fair dies. Consider two events A & B defined by A = sum of numbers on the dies is 6 or 8. B = atleast one dice shows four.

Soln] Sample space

$A = \{(1,5), (5,1), (2,4), (4,2), (3,3), (2,6), (6,2), (3,5), (5,3), (4,4)\}$

$B = \{(1,4), (2,4), (3,4), (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (5,4), (6,4)\}$

$P(A) = \frac{10}{36}$, $P(B) = \frac{11}{36}$, $P(A \cap B) = \frac{3}{36}$

$\therefore P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{3}{11}$

Eg] A box contains 4 bad & 6 good tubes. 2 are drawn out of the box at a time. One of them is tested & found to be good. What is the probability that the other one is also good?

Soln] A → first tube is good
B → second tube is good.

To Find: $P(B|A) = \dfrac{P(B\cap A)}{P(A)} = \dfrac{{}^6C_2/{}^{10}C_2}{{}^6C_1/{}^{10}C_1} = \dfrac{5}{9}$

Eg] A box of light bulbs contains 30 bulbs of which 5 bulbs are defective. If 3 of the bulbs are selected at random & taken out from the box in succession w/o replacement. Find the probability that all 3 bulbs are defective.

Soln] A → 1^{st} bulb is defective
B → 2^{nd} "
C → 3^{rd} "

$P(A\cap B\cap C) = P(A)\,P(B|A)\,.\,P(C|B\cap A)$

$= \dfrac{5}{30} \times \dfrac{4}{29} \times \dfrac{3}{28}$

$= \dfrac{1}{406} = 0.00246.$

Eg] The prob that the management trainee will remain with a company is 0.6. The prob that an employee earns more than 10,000 Rs per month is 0.5. The prob that an employee is a management trainee who remained with company or who earns more than 10,000 Rs a month is 0.7. What is the prob that an employee earns more 10,000 Rs per month given that he is a management trainee who stayed with the company.

Soln] $P(A) = 0.6$

$P(B) = 0.5$

$P(A \cup B) = 0.7$

$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{0.4}{0.6} = 0.667$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$0.7 = 0.5 + 0.6 - P(A \cap B)$

$0.4 = P(A \cap B)$

Eg] A box of 100 gaskets contains 10 gaskets with type-A defect, 5 gaskets with type-B defect & 2 gaskets with both kind of defects. Find the probability that:-

i) A gasket to be drawn has type B-defect under the condition that type A defect.

ii) A gasket to be drawn has no type B defect under the condition that it has no type A defect.

Sol^n] E_1: gasket has type A defect
E_2: " "B" "

1) $P(E_2|E_1) = \frac{P(E_2 \cap E_1)}{P(E_1)} = \frac{2/100}{10/100} = \frac{1}{5}$

2) $P(\overline{E_2}|\overline{E_1}) = \frac{P(\overline{E_2} \cap \overline{E_1})}{P(\overline{E_1})} = \frac{1-P(E_2 \cup E_1)}{1-P(E_1)} = \frac{1-0.13}{1-0.1} = 0.967$

Eg] There are 12 balls in a bag. 8 red & 4 green. 3 balls are drawn successfully w/o replacement. Find the probability that they are alternatively of same colour.

Sol^n] $P(RGR) = \frac{8}{12} \times \frac{4}{11} \times \frac{7}{10} = 0.1696$

$P(GRG) = \frac{4}{12} \times \frac{8}{11} \times \frac{3}{10} = 0.072$

$\therefore P(RGR) + P(GRG) = 0.242$

* BAYES THEOREM

Prior Prob → New Info → Apply Baye's Theorem → Calculate Posterior Prob
$P(A_i)$ → E → → $P(A_i | E)$

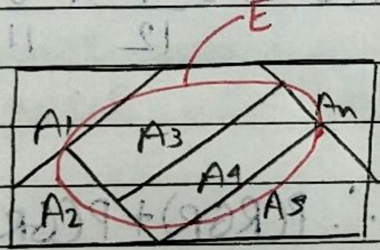
PROOF [10M]

If $A_1, A_2, \ldots, A_n$ are mutually disjoint events with probability of A_i $[P(A_i)] \neq 0$.

Then for an arbitrary event 'E' which is a subset of union of A_i $[\cup_i A_i]$ such that prob$(E) > 0$

$$P(A_i|E) = \frac{P(A_i)\, P(E|A_i)}{\sum [P(A_i)\, P(E|A_i)]}$$

i) $A_1 \cap A_2 \cap A_3 \ldots\ldots = \phi$



$A_1 \cup A_2 \cup A_3 \ldots\ldots = S$

$E \in \bigcup_{i=1}^{n} A_i \; (\cup_i A_i)$

$\therefore E \subset S$

$E = E \cap \left(\bigcup_{i=1}^{n} A_i\right) = \bigcup_{i=1}^{n} (E \cap A_i)$

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ii) $P(E) = P\left(\bigcup_{i=1} (E \cap A_i)\right)$

$= \sum P(E \cap A_i)$

$= \sum P[P(A_i)\, P(E|A_i)]$

iii) By definition of conditional Probability.

$$P(A_i|E) = \frac{P(A_i \cap E)}{P(E)}$$

Prior Probability

$$P(A_i|E) = \frac{P(A_i)\, P(E|A_i)}{\sum P[P(A_i)\, P(E|A_i)]}$$

Posterior Probability

Likelihood

I) The probability $P(A_1), P(A_2), \ldots$ are termed as prior probabilities because they exist before we gain any information.

II) The probability of $P(E|A_i)$ are called likelihood because they indicate how likely the event E occur given each & every prior probability.

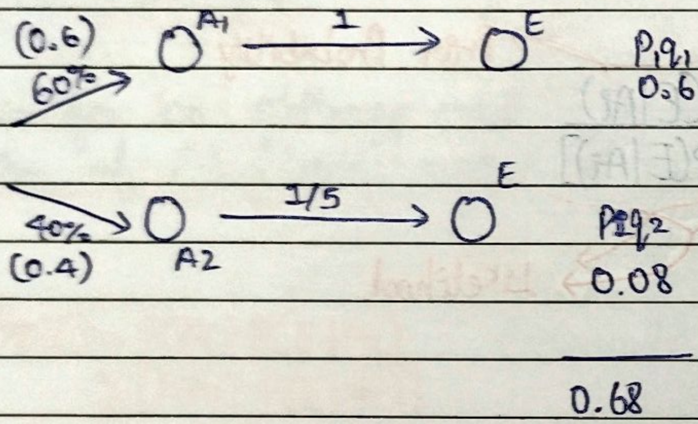
III) $P(A_i|E)$ are called posterior probability because they are determined after the results of experiment are known.

$P(E|A_2) = \frac{P(E \cap A_2)}{P(A_2)}$

$= 0.12$

Eg] A student knows only 60% of the questions in a test each with 5 answers. He simply guessed while answering the test, what is the probability that he knew the answers to a question, given that he answered it correctly?

Soln] $A_1 \rightarrow$ He knows the answer
$A_2 \rightarrow$ He guessed the answer
$E \rightarrow$ He answered it correctly



$$P(A_1|E) = \frac{0.6}{0.68} = 0.882$$

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Eg] The probability of X, Y, Z becoming managers is $\frac{4}{9}, \frac{2}{9}, \frac{1}{9}$ respectively. The probability is that the bonus scheme will be introduced if X, Y, Z becomes manager are $\frac{3}{10}, \frac{1}{2}, \frac{4}{5}$ respectively. If the bonus is introduced what is the probability that the manager appointed was Y?

Soln] $A_1 \rightarrow$ X being manager
$A_2 \rightarrow$ Y being manager
$A_3 \rightarrow$ Z being manager.
$E \rightarrow$ Bonus introduced.

$\xrightarrow{4/9} A_1 \xrightarrow{3/10} E$ — $p_i q_i$ = 2/15

$\xrightarrow{2/9} A_2 \xrightarrow{1/2} E$ — 1/9

$\xrightarrow{1/9} A_3 \xrightarrow{4/5} E$ — 4/15

Total: 23/45

$$P(A_2|E) = \frac{1/9}{23/45} = 0.215$$

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Eg] From a bag containing 3 white & 5 black balls, 4 balls are transferred into an empty bag. From this bag a ball is drawn & is found to be white. What is the prob that out of 4 balls transferred, 3 are white & one is black?

Soln] $A_1 \longrightarrow 4B$
$A_2 \longrightarrow 3B \quad 1W$
$A_3 \longrightarrow 2B \quad 2W$
$A_4 \longrightarrow 1B \quad 3W$

$E \rightarrow$ Drawing a white ball from second bag.

$$\frac{{}^3C_0 \cdot {}^5C_4}{{}^8C_4} \longrightarrow A_1 \xrightarrow{0} E$$

$$\frac{{}^3C_1 \cdot {}^5C_3}{{}^8C_4} \longrightarrow A_2 \xrightarrow{{}^1C_1/{}^4C_1} E$$

$$\frac{{}^3C_2 \cdot {}^5C_2}{{}^8C_4} \longrightarrow A_3 \xrightarrow{{}^2C_1/{}^4C_1} E$$

$$\frac{{}^3C_3 \cdot {}^5C_1}{{}^8C_4} \longrightarrow A_4 \xrightarrow{{}^3C_1/{}^4C_1} E$$

$$P(A_4|E) = \frac{p_4 q_4}{p_1 q_1 + p_2 q_2 + p_3 q_3 + p_4 q_4}$$

$$P(A_4|E) = \frac{p_4 q_4}{\text{Total}} = \frac{3/56}{3/8} = \frac{1}{7}$$

Eg] The contents of urns I, II, III are as follows.

I white, 2 black & 3 red balls → I
2 white, 1 black & 1 red ball → II
4 white, 5 black & 3 red balls → III

One urn is chosen at random & 2 balls are drawn from it. They happen to be white & red. What is the prob that they come from urn III.

Soln] $A_1 \longrightarrow$ Urn I
$A_2 \longrightarrow$ Urn II
$A_3 \longrightarrow$ Urn III

E → ~~One urn is chosen~~ White & red ball are drawn

$$\xrightarrow{1/3} (A_1) \xrightarrow{\frac{{}^1C_1 \cdot {}^2C_0 \cdot {}^3C_1}{{}^6C_2} = 1/5} (E)$$

$$\xrightarrow{1/3} (A_2) \xrightarrow{\frac{{}^2C_1 \cdot {}^1C_0 \cdot {}^1C_1}{{}^4C_2} = 2/6 = 1/3} (E)$$

$$\xrightarrow{1/3} (A_3) \xrightarrow{\frac{{}^4C_1 \cdot {}^5C_0 \cdot {}^3C_1}{{}^{12}C_2} = 2/11} (E)$$

$$P(A_3|E) = \frac{p_3q_3}{p_1q_1 + p_2q_2 + p_3q_3} = \cancel{\frac{2/11 \times 1/3}{\frac{1}{3}\times\frac{1}{5} + \frac{1}{3}\times\frac{2}{15} + \frac{1}{3}\times\frac{2}{11}}} = \frac{2/33}{118/495}$$

$$= \frac{15}{59} = 0.254$$

~~DOUBT~~

Eg] A bin contains 3 different types of disposable flashlights. The prob. that type-1 flashlight will give over 100 hrs of use is 0.7 with the corresponding probabilities for type-2 & type-3 flashlights being 0.4 & 0.3 resp. Suppose that 20% ~~prece~~ of flashlights in the bin are type-1, 30% are type-2 & 50% are type-3.

i) Find the prob that a randomly chosen flashlight will give more than 100 hrs of use.

ii) Given the flashlight lasted over 100 hrs, what is the conditional prob that it was a type-2 flashlights.

Soln] A_1: Type - 1
A_2: Type - 2
A_3: Type - 3
E: flashlight lasted over 100 hrs. → Likelihood.

$$\xrightarrow{0.2} A_1 \xrightarrow{0.7} E$$

$$\xrightarrow{0.3} A_2 \xrightarrow{0.4} E$$

$$\xrightarrow{0.5} A_3 \xrightarrow{0.3} E$$

i) $P(E) = \Sigma p_i q_i = 0.41$

ii) ~~$P(A_2 \cap E)$~~ $= P(E|A_2) = \dfrac{p_2 q_2}{\text{Total}} = \dfrac{0.12}{0.41} = 0.292$

~~iii) Find the prob of flashlight~~

Eg] An urn contains 5 balls. 2 balls are drawn at random & found to be white. What is the prob. of all the balls being white.

Soln] A_1: 2 White

A_2: 3 White

A_3: 4 White

A_4: 5 White

$E \rightarrow$ 2 balls drawn are white.

$\xrightarrow{1/4} (A_1) \xrightarrow{\frac{{}^2C_2 \cdot {}^3C_0}{{}^5C_2} = 0.1} (E)$ $\qquad p_1q_1 = 0.025$

$\xrightarrow{1/4} (A_2) \xrightarrow{\frac{{}^3C_2 \cdot {}^2C_0}{{}^5C_2} = 0.3} (E)$ $\qquad p_2q_2 = 0.075$

$\xrightarrow{1/4} (A_3) \xrightarrow{\frac{{}^4C_2 \cdot {}^1C_0}{{}^5C_2} = 0.6} (E)$ $\qquad p_3q_3 = 0.15$

$\xrightarrow{1/4} (A_4) \xrightarrow{\frac{{}^5C_2 \cdot {}^0C_0}{{}^5C_2} = 1} (E)$ $\qquad p_4q_4 = 0.25$

$$P(A_4|E) = \frac{p_4q_4}{\text{Total}} = \frac{1/4}{0.5} = \frac{0.25}{0.5}$$

$$= 0.5 = \frac{1}{2}$$